

ACADEMIC DETAILS				
Degree	Specialization	Institute	Year	CPI/%
B.Tech.	Mechanical Engineering	IIT Gandhinagar	2023-Present	8.33
Class XII	Physics, Chemistry, Maths	Shree Ganesh Junior College, Korhale	2022-2023	81
Class X		Shree Chhatrapati Shivaji Vidyalaya, Kolpe-wadi	2020-2021	85.60

INTERNSHIPS

Research Intern, SRIP, IIT Gandhinagar

[May '25 - June '25]

(Advisor - prof. Vinod Narayanan) | [Report Link](#)

Worked on the topic “Shock Wave Boundary Layer Interaction over a Flat Plate with Protrusion”, focusing on heat transfer analysis in supersonic flow (Mach 4.5).

Studied the correlation between wall heat transfer rate and the height/width of a protrusion at a fixed distance from the leading edge.

Learned mesh generation using ANSYS ICEM CFD and improved proficiency in ANSYS Fluent by running 32 RANS-based simulations with the SST k- ω turbulence model.

Developed a Python interpolation model to predict heat transfer for protrusion heights from 0.07 to 0.1 m and widths from 0.025 to 0.2 m.

Validated results with data from a research paper and ensured mesh independence for reliable and accurate simulations.

PROJECTS

Yield Prediction Estimation in Aircraft Wing Spar

[March '25- April '25]

(Prof. Harmeet Singh, IIT Gandhinagar) | Course project (team of 5) | [Project Link](#)

Analyzed stress distribution in the I-beam structure of an aircraft wing spar using von Mises and Tresca yield criteria.

Derived theoretical expressions for shear force, bending moment, second moment of area, and calculated principal stresses using Mohr’s Circle.

Implemented these theoretical functions in the backend of a Streamlit web application that takes user inputs for material properties, load distribution, and wing dimensions, and generates shear force, bending moment, and stress distribution plots.

Visualized how the I-beam starts to yield under different load intensities using von Mises and Tresca stress graphs based on the theoretical formulas.

Drag Measurement For External Flow Over Objects

[March '25- April '25]

(Prof. Dilip Srinivas Sundaram | Prof. Uddipta Ghosh, IIT Gandhinagar) | Course project (team of 5) | [Project Link](#)

Conducted experimental analysis to measure drag forces on a cylinder, NACA0012, and NACA0024 airfoils using a subsonic wind tunnel.

Calibrated and recorded velocity profiles at inlet and outlet using a digital micromanometer under the assumption of 2D steady flow.

Calculated drag using Reynolds Transport Theorem and velocity profile data; analyzed deviations from theoretical expectations due to experimental setup constraints.

Gained hands-on experience in aerodynamic testing and instrumentation techniques.

CFD Analysis of Flow Over a NACA 0012 Airfoil

[April '25]

(Prof. Dilip Srinivas Sundaram | Prof. Uddipta Ghosh, IIT Gandhinagar) | Course project | [Project Link](#)

Simulated laminar flow over a NACA 0012 airfoil at 0° and 5° angles of attack using CFD.

Calculated lift and drag coefficients to study the effect of angle of attack on lift and drag force.

Visualized velocity, pressure, and streamline contours, and post-processed results to understand flow characteristics.

Set boundary conditions, created mesh, and improved understanding of symmetric airfoil behavior under varying flow conditions.

TECHNICAL SKILLS

Programming Languages: C, C++, Python.

Tools: Matlab, MSC Adams, Ansys Fluent, Ansys ICEM CFD, AutoDesk Inventor, AutoDesk Fusion, Mathematica (Beginner).

Libraries: Numpy, Matplotlib, Pandas, Cantera.

ACHIEVEMENTS

Qualified Round 1 of ABU Robocon 2025 with a score of 95/100 as Vice-Captain of IIT Gandhinagar team.

POSITIONS OF RESPONSIBILITY

Co-founder and Team vice-captain, Team RUDRA (The ABU Robocon Team of IITGN), IITGN

[October'24-Present]

Founded the ABU Robocon team at IIT Gandhinagar and leading its first participation in ABU Robocon 2025. Our team qualified the first round with a score of 95 out of 100.

Core Team Member,Mean Mechanics(The Robotics Club at IITGN), IITGN

[August'24-Present]

Initiated and coordinated projects as a core member of the Robotics Club at IIT Gandhinagar